

The Concept of Analogy

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Translated from the Danish

Det Kgl. Danske Videnskabernes Selskab,

Filosofiske Meddelelser I, 4.

Copenhagen: Bianco Lunos Bogtrykkeri, 1923

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Introduction

1. In this treatise I have set myself the task of continuing the closer investigation of some of the fundamental concepts which, in my book *Den menneskelige Tanke* [The Human Thought] (1910), I attempted to exhibit as the most characteristic for human cognition, such as we know it after its development hitherto. I have thus wished to give a companion piece to the treatise on *Totalitet som Kategori* (Videnskabernes Selskabs Skrifter. 1917) and to the treatise on *Relation som Kategori* (Videnskabernes Selskabs filosofiske Meddelelser. 1920), and it is the concept of analogy that I shall here investigate more closely.

Provisionally, analogy can be defined as likeness of relation between two subject-matters, that is to say, a likeness which does not hold of single properties of those subject-matters or of single parts of them, but of the relation of properties to one another or of parts to one another. There may be a very great difference between two subject-matters' properties or parts, each taken by itself, and yet there may be a likeness in the relation between the properties or between the parts. Analogy is thus a peculiar kind of likeness. It stands in a certain sense further from identity (absolute likeness) than qualitative likeness, composite likeness and coincidence do, which concern only single properties or parts, and which more easily form transitional forms to identity (cf. *Den menneskelige Tanke* § 21). Yet it must be remarked that there is not only an identity of subject-matter, but also an identity of relation, and already Aristotle distinguishes between quantitative analogy (ἰσότης λόγων), proportionality in the strict sense, and qualitative analogy, which may hold, e.g., between geometrical figures or between organs in different living beings. And there is besides an identity of difference, which lies at the base of the formation of an identically varying difference-series (*Den menneskelige Tanke* § 65) and especially of the formation of the identical quality-series that are fundamental

for exact science (number, time, degree, place) (*Den menneskelige Tanke* §§ 76–82).

Both totality and relation I have given a place among the categories, that is to say, among the fundamental concepts, the forms of thought, which at science's present stage play an essential role in thought-work, determining its tasks and its whole direction (*Den menneskelige Tanke* §§ 51–52; 57). Neither Aristotle nor Kant has given analogy a place among the categories, and in *Den menneskelige Tanke* I followed them in this. Yet in that work I was led to lay much weight on the fact that, in contrast to purely formal thinking (logic and arithmetic), analogy plays an essential role in all thinking about real subject-matters. The opposition between identity (identity of subject-matter in logic, identity of difference and identity of relation in arithmetic) and analogy is therefore of great importance for the right apprehension of the nature and the validity of our cognition. But when analogy really plays so great a role, while identity is properly an unattainable ideal outside the formal world of abstractions, it might be warranted to give it a place in the series of the categories; and in the table of categories to be found on the last page of the treatise *Totalitet som Kategori* this has in fact been done. It is entered there among the formal categories, immediately after the concept of identity, because it lies at the base of that reduction of relations of quality which — with the help of points of view of identity — takes place in the stricter development of the concepts of number, time, degree and place. Such series form intermediate members between absolute identity-series and qualitatively different series of subject-matters. They are formed by analogizing abstraction from the latter. It is here that analogy especially shows its importance. It should now be the task of the present treatise to follow the concept of analogy further in its more special importance for human cognition.

I dwell first on the involuntary analogies which make themselves felt in primitive men and in children, but also, at the high points of the life of the spirit, in the genius. They do not come forward as analogies for the consciousness in question itself; that is possible only for the critical consciousness, which has learned to distinguish between analogy and identity and to see that every grounding and proving transition of thought takes place in virtue of identity. Thereafter I go on to investigate analogy from a formal logical point of view, and in the following sections I seek to exhibit analogies between cognition's different functions and between its different domains.

Provisionally it may be remarked that analogy's importance for cognition shows itself

in three different respects.

Its greatest importance it certainly has by working in a scouting, anticipating and spurring manner. Categories in general lay themselves open to view in the questions that are put (*Den menneskelige Tanke* § 52). Analogy may be the actual road to true cognition, even if the analogy's validity is afterwards rejected. One can, after all, draw a correct conclusion from false premisses. Naturally one cannot prove that the conclusion is correct except by finding the right premisses; but it often happens that an assumption stands for us as true, and that precisely thereby it sets us the task of finding out why it is at bottom true. The premisses (the right premisses) are often the last thing to be found; but one would not seek after them if an analogy had not first led us to an assumption and thereby set us a task. The analytic method rests on this, that a task has been set us, and that one then seeks the conditions for solving that task. If the right premisses are then found, one perhaps rejects the analogy that showed the way — not only if it is false, but also if it is unnecessary. The ground for an analogy's being set up will lie in this, that there is a break in a continuous connection which has formed itself involuntarily for us, and which an involuntary need leads us to hold fast as long as possible. Analogy may here lead to an intellectual venture, which is sometimes crowned with success. One may apply to analogy what Pascal says of the imagination in general, that it deceives all the more easily because it does not always deceive (*d'autant plus fourbe qu'elle ne l'est pas toujours*). If it does not deceive, or if it shows itself to be dispensable, it is replaced by a determination of relation which is finally grounded in identity.

But analogy is not merely preparatory. It may be the last bond of union between different subject-matters or series of subject-matters which cannot be reduced in such a way that a single subject-matter or a single series of subject-matters is seen to lie at the base, or in such a way that they are all seen to have their ground in more deeply lying subject-matters or series of subject-matters. A certain rounding-off and survey becomes possible, in the face of such subject-matters, only by the help of analogy. We do not thereby obtain an explanation of why there are different subject-matters or series of subject-matters, but we establish the actual state of the case, namely that at certain points we get no further than to a connection, a synthesis, which builds only on the remotest degree of likeness — which analogy is. It will appear that both in the face of cognition's different functions (see ch. III) and in the face of its different domains (see ch. IV) we are

finally placed in this way. We end then in open questions, since it remains undecided whether a further reduction, a more deeply lying continuity, might be found, or whether there are here, in the nature of things, differences which are at any rate irreducible for us. Questions can thus be not only the beginning of wisdom, but also its end.

It will have its interest, finally, to dwell somewhat on analogies whose importance lies in their being expressions of poetic and religious moods — expressions which may recall now the analogies of involuntary mental life, now the seeking and finding of scientific thinking. Especially will intimations or hopes of a unity and a continuity lying deeper than experience can exhibit find expression in this kind of analogy.

Chapter I

Involuntary Analogies

2. As already remarked, analogy does not always presuppose that a distinction has expressly been drawn between different subject-matters, or respectively between their properties or parts, and that afterwards, in a certain opposition to the differences, a connection is brought about between the subject-matters on the basis of a likeness of relation. In involuntary mental phenomena, such as sense-intuition, memory and imagination, it cannot always be shown that the elements are first present as separated and only afterwards enter into combination. The importance of these involuntary mental processes lies precisely in this, that they bring about wholes, combinations, which can be analysed only afterwards. Reflection constantly needs a material, but this material need not, as has often been believed, consist in a multitude of independent elements; it may be wholes, which it is then reflection's task to resolve into elements that may not without more ado be regarded as identical with the elements to which the involuntarily formed wholes are due. For newly emerging elements may at once and involuntarily be woven together with consciousness's given content, so that perhaps no alteration at all is noticed. All the same, there is a tendency in reflection to regard the elements which its analysis finds as the whole's original, elementary foundation. There is here an analogy present, which at most can seek its justification in the continuity with which the transition can take place from an involuntarily formed mental continuum to the differentiation to which the very arising of reflection already testifies. Properly, when we speak of the most primitive and involuntary mental processes, all designations fall short, and this is no wonder, since language's psychological designations are fetched from the

clearly differentiating, critically reflecting mental life. The very expression “analogy” we here use in an analogizing manner. It is here as when one speaks of “unconscious inferences”. These are not inferences in the proper sense, but processes, transitions, which can be interpreted in analogy with the proper inferences. The same holds further (as I have already shown in my *Psykologi* and later in *Den menneskelige Tanke* § 5) when we set up the concept of synthesis as an expression for the nature of all conscious life. In epistemology, where we investigate the possibilities of scientific cognition, we must also discuss the possibility that the given might consist of a heap of equally great and mutually irreducible differences. Only an approximation to such a chaotic difference-series can be exhibited (as also to its counterpart, an absolute identity-series); there would in any case be needed a searching and strenuous reflection to establish that such a series really lay before us in a determinate given case.

In the manner in which a single lived experience, from the very moment of its conception, is connected with other simultaneous or past lived experiences, or with traditions into which the individual has lived himself so that these too can be regarded as lived experiences, there lies already an interpretation. The further back we come towards what we call the primitive man’s stage, the closer lived experience and interpretation move to one another, and finally there is no opposition between them to be found. When we nevertheless attempt to hold them apart, we make use of an analogy with what first makes itself felt for testing reflection. And, as has been said, the very concept of analogy is used analogically when we say that primitive men interpret every new lived experience in analogy with what they know from the traditions into which they have lived themselves. — In my treatise *Oplevelse og Tydning* I have sought to describe different ways in which the relation between lived experience and interpretation comes forward historically, as regards ecstatic experiences. —

In his important work *Les fonctions mentales dans les sociétés inférieures* (1910), LÉVY-BRUHL has criticized the formerly common conception that primitive man apprehends everything on the likeness of himself, anthropomorphizes or analogizes. It is, on his conception, not comparison or reflection that is at work here, but an entirely immediate identifying. In primitive man there always lie traditions at the base, with which the new that is lived through involuntarily melts together. The representations handed down from the tribe make themselves felt with such strength of feeling that no need

for reflection arises. What deviates in high degree from what has been handed down is provisionally not noticed; only later is an interpretation perhaps thought out that agrees with the tradition. When the savage sees an image of an object he knows, the image as image brings him nothing new, for the image is the object itself, however he may afterwards arrange this relation for himself. And with all the logical consistency which already distinguishes this stage — which Lévy-Bruhl has wrongly called the “prelogical” — the savage therefore fears to become dependent on one who has an image of him, and he fears that he will come to suffer want of game when a European has drawn or photographed many wild animals in the district and has departed with the pictures. And as it goes with images, so it goes also with names and with ceremonies: they are what they mean. Images, names and ceremonies are indeed only a part of a whole which has other properties besides those that are depicted or named; but this part *is* the whole. Lévy-Bruhl finds in this identifying of part and whole — what he calls “loi de participation”¹ — the most characteristic peculiarity of the primitive mental life. It is certain that the interpretation which primitive man institutes under the influence of a tradition embraced with heartfelt adherence and reverence cannot follow the principles of formal logic. These make themselves felt only when doubt has been awakened; but he is accustomed to regard the tradition as a self-evident foundation for his apprehension of everything that happens. The basis of interpretation is not the work of reflection. But it may lead to misunderstanding when Lévy-Bruhl declares (p. 425) that the collective representations governed by the “law of participation” are “indifferent to the logical law of contradiction”. A distinction must be drawn between the given basis of interpretation (here: the tradition held fast in reverence, which is once for all stamped by the “law of participation”) and the interpretation of single phenomena that takes place on the ground of this basis. Reflection works at all times and on all levels of culture on the ground of determinate presuppositions which cannot themselves at once be made an object for reflection. On the basis which primitive man once for all has, it is entirely consistent that he does not understand why the European distinguishes between the image and the thing. When Lévy-Bruhl calls “the primitive mentality” “prelogical”, this fits the basis of interpretation, not the interpretation made on the ground of that basis.² As there will

¹To this corresponds what GRØNBECH (*Primitiv Religion*, p. 8) calls “Helhedsrealisme” [realism of the whole], and which is characteristic for the primitive mental life.

²Dr. ALBERT SCHWEITZER, who works as a physician in the primeval forests of equatorial Africa, writes,

later be occasion to mention, Lévy-Bruhl has himself in another connection maintained, and rightly, that human consciousness, as regards the ultimate basis for judgments and inferences, properly never quite loses its “prelogical” character. There will always be presuppositions present which we do not and cannot bring forward, even in our most logical reflection. — As regards especially the close connection between name and thing, one can find examples of it even today. There is a tendency to regard the terminology one uses oneself as the only “right” one, and the language one speaks oneself seems to be one with the thing, while foreign languages’ designations often seem foreign to the things. It is told of a German who visited Paris that he reported home that there they called bread “pain”, and added, “Wir nennen es Brot, und es ist doch auch Brot!”

Tradition comes forward for the savage as a closed whole, within which he has no ground for making any distinction. There provisionally rules an immediate belief in the validity of the representations handed down, just as in the sense-perceptions that immediately appear. And the expression “belief in” is even too reflected, for no distinction at all is drawn between the belief and that which is believed in. The very concept of reality or validity is first formed when doubt has made itself felt. The fundamental proposition in accordance with which thinking provisionally goes (though it is of course not formulated) can be expressed by the question “Why not?” Why not rely on everything that comes forward for us with living force as sensation or as tradition? Only later does one learn to ask “Why?” — and reflection can then even reach the point of directing this question against the very thing that has hitherto formed the undisputed basis. For that it is required that new lived experiences should appear so numerous, and with such a peculiarity deviating from everything customary, that they can gain power over what one has already lived oneself into. ALBERT SCHWEITZER declares, as the result of several years’ living together with negroes in Central Africa: “One must not believe that one has got to the bottom of the negro’s world of thought if one has got no further than to look at his superstitious representations and his inherited rules of law. Innermost in him there lives a dark intimation that the apprehension of what is good must come by way of reflection” (*Mellem Floder og Urskove* p. 136). The self-sacrificing physician, who through historical criticism had learned to distinguish between the Christian dogmas and the

after several years’ stay among the negroes: “The natural man thinks more deeply than one commonly believes. Even if he can neither read nor write, he yet institutes reflections to a far wider extent than we suspect.” (*Mellem Floder og Urskov*. Danish transl. 1922. p. 134.)

ethical idealism that comes forward in the Sermon on the Mount and in the parables, experienced this especially when he sought to explain his religious apprehension to the negroes. Yet he experienced also that, alongside the “morality of the heart” that could be called forth in them, there long maintained itself a desire to lie and to steal, and a need for revenge which according to their own traditions was their right and perhaps even their duty.

I must here interpolate the remark that when in my *Psykologiske Undersøgelser* (p. 24 f.) I say that in the relation between new lived experiences and practised, customary representations the chief weight lies on the new, which “sets the whole process going”, this presupposes another stage of development than the undisturbed primitive mentality, where the new so easily glances off against the customary and is able to gain power in the mind only when it comes forward in a particularly insistent manner. —

If one asks after the origin of the system of collective representations that makes itself felt in the savage and determines his interpretation of what meets him, Lévy-Bruhl answers that it “is lost in the night of time”. And since every human group hitherto met with comes forward with certain collective representations, we cannot exhibit any absolutely first beginning. We must here, as with the question of the origin of language,³ content ourselves with analogies fetched from the levels of culture that lie nearest to us, and these analogies must of course be used with great caution. We see that on the levels of culture we know, the traditions that lie at the base of each new generation’s spiritual development undergo constant alterations, additions and displacements, which for an essential part are due precisely to the new generation’s own experiences and reflection. The collective is constantly altered under the influence of the individual. We cannot, as was often attempted in earlier times, think of a time when there was nothing collective, or when what collective there was was due exclusively to single individuals’ experience and reflection. But just as little can we think of a time when minds were governed exclusively by common traditions, without individual experiences and thoughts exercising any influence whatever. It must be assumed that on all levels of development there has taken place a reciprocal action between the common and the individual, and

³Cf. OTTO JESPERSEN: *Language, its Nature, Development and Origin*. (1922). p. 416 f. on the method in the discussion of the question of the origin of language. One must (as already, in their way, Madvig and Whitney had maintained) start out from the manner in which language actually develops, that is, from its history, and from there infer back to a more primitive language than any of those we know.

the difference of the levels can consist only in this, that on earlier levels the common, on later levels the individual, has the greatest influence — yet so that there is again and again a swinging to and fro between the different factors' predominance. Even on the most primitive levels we know, individual differences can make themselves felt and deviations from the tribe's belief come forward.⁴ And the examples of such things would undoubtedly be still more numerous if we had more knowledge of life within primitive societies than we have. We should have found remarkable deviations and initiatives which now are drowned for us in the ruling traditions.

Primitive conditions of life present great uniformity, and therewith great community, in the experiences which the single individuals can make. And on account of the constant living together in group and family, the single man's special experiences and inventions will very quickly become common property, and his name will be forgotten. Added to this, the practical demand constantly makes itself felt; action is required, and life is for a great part lived as reflex-life and instinct-life. Every representation gets its essential importance by releasing movement. But for that it is not necessary to enter into the whole connection to which the representation's content belongs. A single part or a single property is enough to release action, and one need not distinguish between part and whole; the part stands involuntarily as a sign of the whole. The image, the name or the sign works as if the object itself were given in it. The instinct has enough in an intimation to begin to stir. In the fact that a part can thus have the same influence as the whole, and that it may even be advantageous that time and force are not expended on investigating whether the whole is really represented by the single part or property, there might lie an access to an understanding of the "law of participation" or the "realism of the whole". And when a part thus shows itself again and again, in the race's experience, to have the same practical importance as the whole, one sees the possibility that there may be formed a circle of collective representations which obey the law of participation, and which it may be difficult for new lived experiences to break.

Purely logically, this identifying of part and whole (*pars pro toto*) can be formulated as an inference in the second figure: the part releases a certain movement, which the whole would release; therefore part and whole are the same! And this is precisely one of the ways in which an inference from analogy can be formulated. When an instinctive

⁴See my treatise *Sociologi og Religionsfilosofi* (in the book *Oplevelse og Tydning*. 1918) p. 131–138.

action can lead astray, this hangs together with the fact that the “inference” in question is wrong. Only within quite determinate limits, which the consciousness governed by an unconditional impulse to action does not give itself time to establish, can it do to identify part and whole. Neither instinct nor prelogical mentality can therefore maintain itself longer than the inference from analogy practically suffices. Doubt will be awakened when this is no longer the case. A closer psychological investigation will be able to elucidate this relation more nearly and thereby cast light on the involuntary analogies.

3. Just as the adult European is too inclined to read his own developed and reflecting mental life into the mental life of the so-called primitive men, so there is in general a tendency to read functions that presuppose a certain development of reflection into the most immediate and involuntary mental processes. One then speaks of association of representations and of judging, where in reality it is more elementary processes that are going on. This comes in part of the fact that language’s expressions for mental life are essentially fetched from the developed, differentiated mental life, and one is then not clear that it can be warranted only by way of analogy to use such expressions of the most elementary psychical processes. Some years ago (*Psykologiske Undersøgelser*. Vid. Selsk. Skr. 1889) I had occasion to draw attention to this, and I now bring forward a few single points from that investigation, especially in order to show that what Lévy-Bruhl has exhibited for the mental life of primitive men is found again in mental processes that go on in all of us to this day.

a. It has not been sufficiently noticed that there are several different kinds or forms of recognition. In the treatise just mentioned (p. 35 f.) I thought I had to distinguish at least ten kinds of recognition. It leads to confusion when one speaks, as is commonly done, of recognition quite in general and yet in fact keeps to a single form — and that a form which presupposes a certain differentiated development of the life of representation. There is then a failure to recognize the peculiarity of the elementary, involuntary mental life.

The most elementary form of recognition that can be exhibited is as immediate and as involuntary as a sense-sensation can be. A subject-matter here stands for us at one stroke as familiar, with a quality of familiarity, just as a sensation has its sense-quality, without any association, comparison or judgment being demonstrable. A name, a facial expression, a shade of colour can appear with a quality of familiarity without our being

able to remember that we have observed the same subject-matter before, and without our being able to refer it to any determinate connection — that is, without knowing who bears the name, and so on. Perhaps the quality of familiarity gives us occasion to form the judgment that we have heard the name before, and so on. But that judgment then comes precisely afterwards. And if one will conceive such immediate recognition as association, one must formulate it as a limiting case and think of the matter thus: that when a sensation of sight or of sound is repeated, it can, in the very moment of its birth, fuse with the after-effect of earlier sensations of the same character (however one may care to conceive such an after-effect). There takes place, then, a momentary reproduction, and what is reproduced fuses at once with the sensation that is coming into being. The quality of familiarity can then theoretically be conceived as the product of two elements. Physiologically regarded, we have here an example of practice. All practice of a function is due to a fusion of a new release with the after-effects of earlier releases.

This exhibiting of immediate recognition as an elementary phenomenon, lying on the boundary between sensation and memory proper (I have called it bound memory), was not acknowledged or attended to when it first appeared. It was rejected by Danish psychologists, and writers such as Wundt, Meinong and Bergson assumed without more ado that association, comparison or judgment is always the condition of recognition. It has interested me the more to see that G. E. MÜLLER, who is an authority as regards the doctrine of representations, in his psychological lectures printed as manuscript (1913) § 26 describes the same phenomenon that I call immediate recognition, under the name of indeterminate recognition. “Recognition of an object of observation is,” he says, “indeterminate when, in observing the object, we merely have the impression that we must somewhere have observed it before. When the object not only comes forward for us as familiar, but in addition awakens the recollection of its name, of the circumstances under which we observed it earlier, and so on, then we have a determinate recognition.” I have against this description only this to object, that the quality of familiarity need not be bound up with a consciousness of (“the impression of”) our having apprehended the object before. It can be without consequence. It is already an inference when we take it that we have apprehended the object before — an inference drawn from the difference between sensations with and without the quality of familiarity, an immediate inference of course. When immediate recognition at once releases a reflex or an instinctive action,

no such inference is drawn, and “the impression” will not make itself felt.

MÜLLER finds (in a handwritten addition to the printed lecture) the explanation of indeterminate recognition in this, that after-effects of earlier observations, and perhaps indistinct reproductions of these, are released by the new sensation; or that subject-matters we have apprehended earlier are, on repetition, apprehended more easily and more quickly than before. In quite the same way I had also referred to the law of practice, which finds application both to the psychological and to the physiological side of the phenomenon. It can of course often be difficult to decide whether, and in what degree, an indistinct association of representations takes place. I have therefore designated immediate recognition as a limiting case, and I have in my *Psychologi* compared the relation between immediate recognition and association of representations with the relation between the tangent and the secant: the nearer the secant's two points of intersection with the circle come to lie to one another, the more does the secant approach the tangent. Repeated and exact observation has shown me many degrees here, and there is every occasion for closer investigation.

Even before MÜLLER's lectures were printed, BETZ (*Archiv für die gesamte Psychologie*. 1910, p. 267) had declared: “Erst die vielzitierte Abhandlung von Høffding über Wiedererkennen zeigte, dass die Dinge bei weitem nicht so einfach liegen, als es den Assoziationstheoretikern schien”. — The Neoplatonist Olympiodorus,⁵ moreover, had already mentioned the phenomenon. In his commentary on Plato's *Phaedo* he supplements his master's doctrine of recollection by drawing attention to the fact that there is not only a transition from one element to another on account of likeness [association by likeness], or from one element to another element different from it with which it has earlier occurred together [association by contiguity], but that also “the same can be connected with the same”. When someone recognizes Socrates, for instance, after having forgotten him for a time, then it is, says Olympiodorus, just as much a sensation as the first time he saw Socrates; in such recognition there takes place no transition of cognition (οὐ γέγονε μετάβασις γνώσεως). That Plato does not mention this kind of recognition, Olympiodorus explains by the fact that in the *Phaedo* he had use only for the free (“determinate”) recognition, in which, according to him, “the Ideas sprang forth in the soul”.

⁵ *Olympiodori in Phædonem Commentaria*, ed. W. NORVIN. 1913. p. 68.

b. When there lies before us a composite subject-matter whose elements (parts or properties) we have often apprehended in succession, we need not always go through all the elements anew for recognition to take place. Successive recognition can be replaced by partial recognition. If the subject-matter's elements are *a b c d*, it can be enough that *a* is recognized. We have here quite the same phenomenon that LÉVY-BRUHL has designated as participation. Partial recognition has great practical importance, since it would often take too long a time if the whole series of elements had to be recognized before we were sure of having the subject-matter before us, so that we could proceed to act towards it. The subject-matter thus contains more than we notice, but we confine ourselves to the strictly necessary. We recognize, e.g., a man by his walk or by some peculiarity of his dress, a chemical substance by its smell, and so on. As with immediate, indeterminate recognition, it is the need for action that shortens the process. The necessity of intervening, if the recognition holds good, weighs practically more than the certainty that the apprehension is correct. When spontaneous or instinctive action leads into misfortune, it is because the single mark does not hold good.

Partial recognition can become sense-illusion when the immediately recognized element has in different cases been connected with different other elements. When both the series *a b c d* and the series *a f g h* have occurred, and I, when *a* comes again, at once rely on *b c d* being there too, I can be surprised by *f g h* presenting themselves instead. I use an example from my *Psykologi*. On a walk along the Jammerbugt on Jutland's west coast it happened to me, a whole day through, that I took pieces of wreckage sticking up out of the sand for men. There took place here an involuntary supplementing of what was really seen. The image that involuntarily formed itself for me contained more than proved to be given. The supplementing here takes place just as involuntarily as in the well-known experience with the blind spot, or as when, after a part of the lower lip has been locally anaesthetized, one feels a hole or a depression in the rim of the glass one drinks from. Such a supplementing lies, seen from our modern point of view, at the base of primitive man's participation. The "prelogical", which only by a questionable analogy can be described as having arisen through association or through judging, we thus know from daily experience. Purely logically, sense-illusion can be formulated as a fallacy; but it is all the same unwarranted on that account to read such an inference, as an actually occurring psychical process, into the elementary phenomena we here have to do

with. We simply have no right to identify the elements which analysing reflection finds with those that co-operate in the original lived experience. Here the analogy between psychological analysis and chemical analysis fails.

When one formulates the sense-illusion, or partial recognition in general, as an inference, it is the second or the third figure that we get. The property *B* is found both in the subject-matter *A* and in the subject-matter *C*; from this is inferred partial or absolute identity of *A* and *C*. (Second figure.) Or: the subject-matter *B* has both the property *A* and the property *C*; from this is inferred that there is partial or absolute identity between *A* and *C*. (Third figure.) As we shall see, we have here two of the ways in which an inference from analogy can logically be formulated. What is questionable in every inference from analogy already comes forward clearly here. —

In all participation, that is, in immediate, partial and illusory recognition, an involuntary need for whole-formation makes itself felt. Even if only a single element is actually given, one at once makes it into a whole. GRØNBECH has therefore rightly designated what Lévy-Bruhl calls participation as realism of the whole. Such an involuntary impulse toward the whole is especially characteristic of the first stages of development of conscious life, which bear the stamp of naivety; it does not — fortunately — disappear later either, though it becomes an object of analysis and criticism. It lies at the base of what I have called concrete and practical intuition. By concrete intuition I understand sense-intuition, memory or imagination. In all these three forms, which do not from the first stand as different in principle from one another, a whole is built up on the occasion of a single element or of a few elements. Since it will often be a ruling feeling that determines the centre about which the whole-formation takes place, concrete intuition is more or less akin to practical intuition, in which all the mind's forces can work along, without there needing to be any consciousness of the sources of the whole-image or of the energy with which it is formed and held fast. All practical faith, indeed all personal concentration upon a thought or a task, is of this kind. Here too we are reminded of the "prelogical" consciousness, which according to Lévy-Bruhl has a purely emotional and mystical character, since it is the tribe's sacred traditions that decide how the individual interprets, or rather integrates, what he lives through.⁶

⁶Cf. my treatise on *Begrebet Intuition* (Vid. Selsk. Oversigt 1914) p. 75–77 and my work *Henri Bergson's Filosofi* p. 22 f. — In contrast to the two kinds of intuition here mentioned I set analytic and synthetic intuition; the former expresses reflection's testing, the latter its gathering function.

c. In all recognition, likeness to what has earlier been lived through plays a role, and to that extent recognition can be regarded as a limiting case of association by likeness. As regards association by likeness in general, there has been dispute whether any such thing exists at all. My position on this question has, ever since the first edition of my *Psykologi*, been that I do not assume any pure and unmixed association by likeness, since in all demonstrable examples of likeness there will always be certain differences present. No one has been clearer about this than Plato — it is of course for that reason that he makes perfect likeness an “Idea”. And the elements of difference will be able to release association by contiguity, which plays a greater or a smaller role in the transition to the following representation. Every actual association will therefore be a mixture of association by likeness and association by contiguity, and the discussion must in each single case turn on the degree to which moments of likeness co-operate. My formula for association by likeness is therefore $a_1 + a_2$, and the indices designate the differences which always make themselves felt, even where the likeness is preponderant. Since it is peculiar to such transitions in the life of representation as are essentially due to likeness that they take place involuntarily, and especially when the mind is strongly occupied and moved, it will be difficult to exhibit such transitions experimentally; it is very difficult in psychological experiments to avoid preparation and expectation, whereby association by contiguity is at once favoured.⁷

While many psychologists (especially Danish ones) took a quite dismissive attitude towards the assumption of an association by likeness, even with the limitations stated, G. E. MÜLLER, who has quite especially investigated connections of representations, expressed himself more cautiously. He did not find it probable (in the lectures mentioned above) that there was pure association by likeness; when a portrait calls forth the representation of the person himself, it may happen by its first calling forth the name, and this then the person. There is of course no doubt that it can go this way. I have, however, not been able to convince myself that it fits all cases; I can well recognize a portrait even if I have forgotten the name, which perhaps even after the recognition I have trouble in recalling.

In the same year in which Müller had his lectures printed as manuscript, he took part as a subject in some experiments conducted by his pupil Louise Schlüter, which were to test the relation between two methods of teaching a foreign language, of which the

⁷See, besides my *Psykologi: Psykologiske Undersøgelser* (1889), p. 43–59.

one builds on a direct connection between an object and the foreign word (the intuition method), while the other first names the domestic name of the object and lets that recall the foreign word. It showed itself here that when, the day after an object had been shown and the foreign word learned, a similar object was shown, the foreign word came up without any support from the domestic word. This presupposes that the object seen on the preceding day was reproduced from the object now seen, although it was not quite the same. This Louise Schlüter regards as proof that there can be a real association by likeness,⁸ and Müller has declared himself in agreement.⁹

I shall only remark on this, that in observations and experiments on this relation a distinction must be drawn between the different kinds of likeness. In Louise Schlüter's example it was, so far as I can see, qualitative likeness that was in question: the object shown on the second day was not the same as the one seen on the first day, but "ein ähnliches Exemplar derselben Art". It was thus not what I call coincidence that made itself felt here. But since the relation between examples of the same species is for the most part likeness of relation — inasmuch as it need not be a likeness with respect to a single part or property that is decisive for the species-concept — it would be worth investigating how, in cases like Schlüter's, qualitative likeness and likeness of relation stand to one another.

At bottom, Schlüter's case is not different from the cases on which MÜLLER in his lectures (§ 20) has built the form of association which he calls active substitution, and which is an association by likeness based on an association by contiguity. When a is associated with the representation b different from it, then a representation α which resembles a can also call forth b , whether the relation between a and α is qualitative likeness or analogy.

Such active substitution is of great interest for our investigation of analogy in its epistemological respect. For when it shows itself that a and α practically play the same role, whatever kind of likeness it may be, we see here a psychological possibility for that confusion of identity and analogy which proves so fateful in the history of thought. There will from the first be no doubt about the validity of the transition from α to b . Practically

⁸*Beiträge zur Prüfung der Anschauungs- und der Übersetzungsmethoden* (Zeitschrift für Psychologie. 1913) p. 73; 103.

⁹In handwritten additions to a copy of the lectures, which Professor Axel Herrlin has kindly placed at my disposal.

regarded, there is a tendency to treat all likeness as identity. This is seen already from the much-discussed phenomenon that a burnt child dreads the fire. According to Stuart Mill, we have here merely an association by contiguity. But the child provisionally shuns all objects that resemble the one on which it once burnt itself.

4. In the child's play and language, in artistic intuiting and in the genius's working, involuntary analogies make themselves felt which are as little as those discussed in the foregoing due to express comparison or inference, and are not due either to an association in which every single member is apprehended by itself and only thereafter connected with other members.

a. The child's play is often due to imitation of the grown-ups' doings, but the child can immerse itself in it with so complete a living-into that the difference between what the play "is" and what it is to "mean" does not make itself felt at all. If the play is interrupted from outside, this can awaken pain and indignation. A child was playing housewife and had laid its play-table with various imagined dishes. When it now became the house's dinner-time, and the child was wanted to sit down at the real table with the rest of the family, it became violently incensed and burst into tears, "for a mother doesn't just run away like that from the table she has to look after". The indignation shows how deep the living-into the play had been. In a similar way it can go with the spectator at a drama, when he is suddenly disturbed in his full merging into what he sees and hears. It has wrongly been attempted to explain the attraction of a tragic drama by supposing that the reader or spectator is conscious of standing outside what is presented. Such a knowledge would make the full absorption impossible. The absorption presupposes that we for a while identify our own fate with the one presented. It is an illusion, a "participation", which is perhaps lifted when the curtain falls or the book ends, but whose resonance or "perseveration" can make itself felt for a long time, since we may have a tendency to live again and again in the same mood as when we were immediately under the drama's influence.

Outside play, too, children show a capacity for analogical lived experiences. A state or a situation that has brought them satisfaction they will have continued, either in such a way that they themselves are still part of it, or, more disinterestedly, so that the state is merely continued after the same scheme as before. In both cases it is equivalents that are sought. AVENARIUS has, in his interesting principal work *Kritik der reinen Erfahrung*,

sought to show that here — in the need to continue a state, even if one thereby goes beyond the old forms — a universally human tendency makes itself felt. He adduces the following amusing example from child psychology (which decidedly does not belong among the disinterested ones). A little boy is sitting comfortably enough in an armchair and letting his guest, a little girl, stand. His mother reproaches him for it, and he vacates the seat, whereupon the girl settles herself in the chair with her toy on her lap. The boy looks at her a little and then proposes that she should lay the toy on the floor and let him sit on her lap.¹⁰ — To the disinterested continuation of a situation, on the other hand, belongs the following example. A little boy was riding on his grandfather's knee, but had to break off the situation when he was to go to bed. A little after, he came rushing back with great eagerness, bringing his toy bear, which he placed on the grandfather's knee. In this way a new member was set into the relation, which otherwise would have had to cease. Perhaps there was also sympathy with the old man, whose lap was now empty. —

In the child's acquisition of language, not only recollection and imitation of the grown-ups' speech, but also involuntary analogizing, plays a role. The child has, e.g., heard the expression "Jacob's hat" together with the corresponding visual image, and now forms on its own account the expression "Peter's hat" on a given occasion, without the general representation "hat" being a necessary intermediate member. Children moreover often form regular verbal forms on their own account, when they have not learned the irregular ones. Afterwards they can then practise using the irregular forms which usage requires, instead of the regular ones they had themselves formed in involuntary analogizing.¹¹

b. The fact that creation by genius was comparatively late acknowledged as something other and more than the product of a mechanical joining-together of particulars in a conscious purpose¹² shows how hard it has been to grasp involuntary psychical whole-formation. All mental processes were regarded from the standpoint of analysing reflection. After Rousseau (especially in the article "Génie" in his *Dictionnaire de Musique*) had described genius as the capacity for involuntary creation, and had entreated "l'homme

¹⁰*Kritik der reinen Erfahrung*² II p. 493, cf. I, p. 136. — AVENARIUS finds the same tendency in such cases as those where men, through sudden blows of fate or through the enfeeblement of age, can continue their spiritual life only by the help of forms of faith which in their vigorous years they had abandoned. *Kritik der reinen Erfahrung*² II, p. 295.

¹¹OTTO JESPERSEN: *Language*, p. 130 f.

¹²Cf., e.g., the way in which HOLBERG himself explains the coming into being of his various works. See my *Mindre Arbejder* II, p. 150–154.

vulgaire” not to misuse this sublime word, it was KANT who fixed the concept in the form in which it is now commonly used, in apprehending genius as that rare originality of natural endowment which makes it possible to bring forth works of exemplary character — works which cannot be imitated, but which can awaken originality in other natural endowments (*Kritik der Urtheilskraft* § 40). Genius is, like instinct, an immediate union of need and capacity. The involuntary and the exemplary together make up the essence of genius. GOETHE rejoiced (*Dichtung und Wahrheit*. 19. Buch) that the best word for involuntary and model-worthy creation was thus rescued. — A fate like that of the word genius threatens the word imagination, when one uses it of all free representation, even where this is merely recollection, and does not restrict it to mean the capacity for the new formation of concrete individual representations. MEUMANN is surely right when he holds¹³ that the narrower meaning of the word imagination first arose under the influence of the modern age’s more penetrating study of artistic production. Artistic imagination leads to works which to an observer can seem to be the fruit of a reflection that with full consciousness lays in the single features in such a way that they co-operate with other features to the formation of a whole. On the occasion of a feature which Solger had brought out in his analysis of the characters in “Wahlverwandschaften”, GOETHE said: “Ich habe selber nicht daran gedacht, aber es liegt darin”. (*Gespräche mit Eckermann*. 21. Januar 1827.)

The artist thus comes involuntarily to say more than he is conscious of. That he analogizes, and thereby comes to connect elements that were otherwise separated, he does not notice until perhaps later, if testing reflection awakes. By a living-into — or, as the Germans say, a feeling-into (*Einfühlung*) — of the subject-matter, which involuntarily leads him to seek and to find expression for its value, he can awaken, if not an independent creation, then at any rate an independent capacity for a living-into like his own. In this latter living-into it is not only the subject-matter’s content, but also the form under which it comes forward in the work, that exercises a decisive effect — just as the artist’s living-into the subject-matter leads him from the first to a forming activity which might let the subject-matter come forward in its whole fullness. In the definition which JULIUS LANGE (*Om Kunstværdi*. 1876) gave of artistic value, as the value the subject-matter has for the artist and which it acquires through his work for the beholder, the form

¹³*Intelligenz und Wille*², p. 136.

of presentation did not come into its own (as I sought to show in a review of Lange's fine book in "Nær og Fjern" 1876); this Lange regarded as something purely external and mechanical. In the artistic genius, need (feeling of value) and capacity (forming power) are joined from the first — and joined in the most intimate way, since there is a need to use the capacity, and not merely to feel the value, and since the feeling of value presupposes a capacity for opening oneself to what can call forth that feeling. FRIEDRICH VISCHER has rightly maintained that the feeling-into must determine both the work's form and its content. Lange, by emphasizing the living-into of the subject-matter, pointed to what is involuntary in the imagination and in the genius; but to this belongs also the involuntarily forming activity. On Lange's definition it might look as if the work of art were merely an external, inessential means by which others than the artist himself could experience the subject-matter's value. Need and capacity are not bound together in so outward a way, even though the carrying through of the forming activity can at many points make possible, indeed require, conscious deliberation and choice.

The concept "Einfühlung" was known neither to Lange nor to me when we conducted our discussion. It had been set up some years earlier by the German aesthete ROBERT VISCHER, but became more generally known only through VOLKELT's and FRIEDRICH VISCHER's writings.¹⁴

While for the primitive consciousness the image stood as the bearer of a magical power which made its possessor master over the subject-matter, in art it is the bearer of a value that was born in the artist's soul and through his work is born again in the beholder's soul. In both places there is an analogy which works in an involuntary manner, and which is seen as analogy only by testing reflection. That the involuntary mental life's manner of working has not been understood led to a failure to recognize the nature both of the genius and of primitive man. But the understanding that can be won of the involuntary mental life comes about itself by the help of analogy, since the understanding starts out from the differentiated life of consciousness, which can make the primitive syntheses clear to itself only by dissolving them.

¹⁴JOHANNES VOLKELT: *Der Symbolbegriff in der neueren Ästhetik*. 1876, p. 55. — FRIEDRICH VISCHER: *Das Symbol* (Festschrift til Zeller. 1887) p. 170–187. — ROUSSEAU knew feeling-into at first hand. His joy in nature springs from it. Cf. *Confessions* (éd. Bever) I, p. 263 f.: Je dispose en maître de la nature entière; mon cœur, errant d'objet en objet, s'unit, s'identifie à ceux qui le flattent. — But in Rousseau the need was greater than the capacity. A work of art he could not form (apart from the work of art his own *Confessions* are). But his heartfelt feeling gave the impulse to modern poetry.

Answering to the image's magical significance there stands, in art, its symbolic significance; but in the history of religion magic and symbolism are broken wholly apart only in the most recent time.

5. The symbol rests on participation, in that a part stands as the expression of the whole and works as such. The small can become a symbol of the great when it presents in itself relations that have a likeness to relations within the great. For the great humour, everything single, however limited it may be, can become a symbol by being seen as part of a great whole and as presenting the same type as the whole. "The humblest thing defends its place in life's garland." But life's garland is never wholly bound, and the possibility is always present that new flowers are to be woven in. To this the great humour will not be blind; it looks humorously on its own garlands,¹⁵ and it agrees with GOETHE (*Gespräche mit Eckermann*. 5. Juli 1827) that in images used as expressions for life there always remains something incommensurable for the understanding. The image's value rests precisely, on one essential side, on its giving as a whole what for testing reflection stands as an unclosed connection of elements — perhaps of constantly changing elements. Seen from this side, the image gives less than the always advancing experience and reflection. As a whole-image it gives more, since as such it has elements not to be found in the subject-matter from which it is fetched. When God is called Father, the element of omnipotence comes in with it, which unfortunately is lacking in every father given in experience, who must often struggle hard both for himself and for his own. Or (to use FR. VISCHER's example) when the lion is a symbol of magnanimity, it has a property the real lion lacks. Pseudo-hallucinations are therefore often better suited as a foundation for symbols than hallucinations proper, which as a rule take in all the elements of reality.

Religious symbolism is especially characterized by the fact that its foundation of feeling has formed itself under the struggle for what, for the human group in question, is the highest thing of value. The images in which expression is found for the experiences made in the struggle for existence are fetched from the great oppositions of nature and of life: light and darkness, life and death, joy and sorrow, power and powerlessness, good and evil; and, as the most elementary nature-religions give place to higher forms of religion, a need makes itself felt to see these oppositions represented and working

¹⁵*Den store Humor*. Chap. 6 (Forstaaelse og Humor).

through more or less individualized beings.

But like the concept of analogy in general, the concept of symbol too is first formed at the standpoint of testing reflection. The difference between part and whole, between the sign and the signified, does not make itself felt from the first. What for the artist and the artistic beholder holds only in moments of enthrallment is, for the positively religious consciousness, so long as it subsists in its full peculiarity, the constant state. Every distinction between part and whole, image and subject-matter, is heretical. Hegel wrongly held that in all symbolic apprehension a dissatisfaction must be felt on account of the ever-present disproportion between form and content. Since Hegel presupposed such a dissatisfaction already at primitive stages, and in general within the positively religious consciousness, it became comparatively easy for him to show that it was a liberation for the religious consciousness to pass over to forms that could subsist for testing reflection; and the difference between religion and philosophy became for him in the end only a difference of form, while the spiritual content was to be the same.¹⁶ Hegel reads the cleavage to which reflection can lead into the religious consciousness itself, for which, on the contrary, the relation between part and whole, sign and subject-matter is a relation of identity. It is denied, e.g., from the side of positive religion, that the expression Father used of God is figurative. So far is it from being experiences of the significance of the human father-relation that are to lie at the base when God is called Father, that the case is precisely the reverse: it is God who is the ground and the model for every father-relation in the world (the Epistle to the Ephesians III, 15). By virtue of this identity it is possible that actions which for reflection stand as symbolic – such as prayer, sacrifice and sacrament – can be assumed to exercise an influence on the Godhead. This is consistent, for by virtue of the relation of identity there is a real connection between sign and object. Added to this, the positively religious symbolism is the result not only of the individual's but of a whole human group's experiences of life, and thereby comes forward for new generations as the given and self-evident expression of what can be experienced about human life and its conditions.

Concepts such as analogy and symbol are correlative concepts; they express a reciprocal relation. As we have already seen with the father-symbol, the relation therefore turns quite naturally round. From the world of experience in which human life is led,

¹⁶ *Den nyere Filosofis Historie*.³ III, p. 182–186.

symbols were fetched for a higher world; but in the end the very world from which the symbols were originally fetched becomes a symbol. In the mystics of the Middle Ages this turn comes forward in a characteristic way. For Hugo a St. Victore the whole bodily world is a symbol of the spiritual world. And when Goethe uses medieval expressions to denote the soaring above everything finite, he declares

Alles Vergängliche ist nur ein Gleichniss.

Medieval thinking moved, as it has aptly been said, as often from symbol to symbol as from experience to symbol. There had once for all been awakened a fundamental mood which constantly sought new expressions, and in relation to which different experiences of life could mean the same. Here the emotional foundation is at work which is in general determinative for all “participation”, all “prelogical mentality”. Only critical reflection makes that demand on religious representations which Schleiermacher maintained, that each single symbol shall be formed by itself on the foundation of a particular lived experience. In the doctrine of faith, he says, the one proposition must not be derived from the other, but is to be set up immediately with reference to a single determinate lived experience, and only then afterwards compared with other, previously established propositions. (Der christliche Glaube § 17.) So long, on the other hand, as the once developed fundamental mood maintains itself, new symbols will spring again and again from it, the one developed in analogy with the other. Thus YRJO HIRN (*Det heliga Skrinet. Studier i den katolska Kyrkans Poesi och Konst. 1909*) has shown in a fine way how the heartfelt worship of the Madonna leads to a series of images which express the fundamental mood from different sides: the dwelling-place of the Host in a shrine on the altar, the place where the divine comes forward in bodily form, became a symbol for the Madonna, in whose womb God became man. But the Madonna is at the same time the morning star that heralds the sun’s coming, and she is the spring, where light and warmth relieve the darkness and cold of winter. The mighty expansion of feeling constantly puts out new shoots.

The involuntary formation of images is the presupposition of all dogmatics. Not all symbols become dogmas. There is talk of dogmas only when certain symbols stand as norms which individuals ought to follow in their attempts to interpret their experiences of life. A choice is then made, and what is chosen stands fast in virtue of the religious

community's authority over its members. The dogma rests on a mixture of lived experience, reflection and authority. From what has been dogmatically fixed a struggle is then carried on against freely developed symbols, as well as against the attempts to apprehend the dogmas themselves as mere symbols. Dogmatics' struggle against symbolism is a struggle between "it is" and "it means".

A free formation of symbols within a religious frame is presented by Jesus' parables, in which through narratives or features seized out of life he makes intuitable the relations and the demands of the promised kingdom of God. An ethical-religious wisdom of life is here uttered in poetic form. The figurative is chosen with deliberate intent, and often (especially in the parables of the Gospel of John) passes over from symbolism to allegory.

The symbol becomes allegory the more expressly a distinction is drawn between the two members in the symbolizing. In allegory the relation between the two members is more external than in the symbol. Poetic symbolism springs, like religious symbolism, from the enthrallment a subject-matter has awakened; but it does not exclude, as religious formation of images in the classical ages of the religions does, that the enthrallment may alternate rhythmically with a clear consciousness of the formation of images that has taken place.

In intuitive natures, where intuition and imagination have the preponderance over reflection, there will be a tendency to live in symbols as long as possible. Still stronger will this tendency be in ecclesiastical natures, for whom it is a need to hold fast to inherited symbols — not only on account of their intuitability, but because they are the inherited expression of the fathers' faith, and thus on account of a feeling of solidarity with the past. They hear a centuries-old tone in the church's hymns and find again the race's dearly bought experiences in the church's dogmas. Yet it must be noted that it need not be intellectual criticism alone that bursts the inherited forms; it may also be a new and strong movement of feeling that makes it impossible to find one's own experiences of life expressed in these forms.¹⁷

¹⁷Cf. my *Religionsfilosofi* §§ 95–96. — *Oplevelse og Tydning* p. 102–107.

Chapter II

Analogy and Logic

6. In the involuntary analogies, likenesses and differences are from the first so woven into one another that it can be difficult to find the boundary where the likeness ends and the difference begins. The definite attempt to find such a boundary, through a sharp determination of the different degrees of likeness (and of difference), marks the beginning of scientific thinking. Plato stands in the history of thought as the first who drew attention to there being many steps upward from the likenesses that come forward for immediate apprehension to pure identity, the “Idea of likeness”, which alone makes strict science possible. What Parmenides had proclaimed, under the name of “the One”, as the sole condition of true knowledge is determined by Plato more exactly as identity. — In my treatise on the Platonic dialogue *Parmenides* (Vid. Selsk. filosofiske Meddelelser Nr. 3) I express myself perhaps too strongly when I say (p. 28) that Plato “expressly declares” unity and identity to be the same; more cautiously it is said on p. 30 that it is “let out” that unity and identity are the same. The fact is that Plato in the strange second part of the dialogue apparently denies this; but since he grounds the denial only on the point that *several* things (τὰ πολλά) can be One (i.e. can come under the same concept) without coinciding, and adduces no proof at all in the case of “the One”, which is just what is in question, I find here an involuntary concession. So much is certain, that Parmenides’ “the One” is obscure and mystical when it does not mean identity — that is, when it does not express at once the fundamental law of all thinking and the highest ideal of all thinking.

For Plato, now, the Idea of likeness was not merely fundamental law and ideal, but the expression of true being, and the degrees of likeness were so many lower and higher

steps of existence. All the same, in the setting up of this Idea a direction was given for the dissolution of the involuntary analogies. When that dissolution is carried out with a hard hand, absolute likeness and absolute difference come to stand in opposition to one another, identity in opposition to chaos, and it becomes cognition's task to find a road between these oppositions. It is characteristic of the sharp Greek thought that not only the Idea of absolute identity but also the Idea of absolute difference is brought forward, albeit with a polemical purpose (by Zeno). And towards the close of his career Plato saw that the principle of identity stood in the end as powerless in the face of the manifold differences that existence presents, and therefore refers to analogizing as a kind of "spurious thinking" (λογισμὸς νόθος. Tim. p. 52 B). Analogy stands here, then, as a last resort, where no higher degrees of likeness can be exhibited. Whereas in mythological thinking it was everything, it now comes to denote the lowest stage of cognition. The programme of European thinking was herewith given. And the task then becomes to exhibit the different intermediate forms between chaotic difference and absolute identity, for it is in these intermediate forms that human thinking actually moves. The order of rank of these intermediate forms (of which I have described six in *Den menneskelige Tanke*) will depend on how far the subject-matters before us can be ordered in such a relation to one another that inference becomes possible. It will then show itself that the more inference becomes possible, the more we approach pure identity. It shows itself at the same time that identity no longer stands as the highest form of existence, but as the form of thought which makes grounded cognition possible without calling in the help of observation at every moment. The principle of identity dawns most clearly in such series as those where each member stands to its predecessor as that one does to its own predecessor. If, e.g., *A* is subject to *B* and *B* to *C*, or *A* is smaller than *B* and *B* smaller than *C*, then *B* can be eliminated and we get a new judgment. The presupposition is *B*'s identity with itself, whether it be set in relation to *A* or to *C*. But such a presupposition holds for every member in every series we set up. To that extent the principle of identity lies at the base of all series-formation; but cognition consists not merely in the formation of single concepts, but in the exhibiting of intelligible relations between concepts, in rational series-formations.

The expression "participation", so happily chosen to denote involuntary analogy, can well be retained after logic has begun to have a word to say concerning the presuppositions

of the course of thought. It denotes a partial likeness which in the involuntary analogizing was made into an absolute likeness, in spite of the difficulties there are in grounding the right to assume, from a likeness at a single point, an agreement in general. It is precisely these difficulties in “participation” that set scientific thinking its tasks.

If one seeks a presupposition which for all that lies at the base both of prescientific thinking’s involuntary analogizing and of the scientific search after grounded transitions, and which lies at the base of all series-formations from the chaotic difference-series to the absolute identity-series, then such a presupposition lies in the law of synthesis, which holds both in empirical psychology and in rational epistemology. It lies in the concept of a series — whether the members are held together by mere memory or by grounded understanding of the transitions — that different members are gathered together with one another, either as different or as alike. If thought constantly lost the preceding members when it went over to the following ones, no series-formation would take place. The more it becomes possible to ground the transition from member to member, the smaller the role that mere memory — which is sole ruler in the chaotic difference-series — needs to play; the transitions then take place in virtue of a law.

The fundamental law of synthesis rules at all stages of the connection of representations, and thereby it becomes possible to compare these different stages and to exhibit the relation between them. If the different sciences, which find themselves at different stages between chaos and identity, did not have such a common fundamental form, we could not set up the very concept of science as common to them all. We can now elucidate not only one science through analogy with other sciences, and one part of a science through analogy with other parts, but also prescientific thinking through analogy with the scientific; and we can within psychology elucidate the elementary mental functions by analogy with the more developed ones. Such analogies are not mere fictions, as VAIHINGER held in his *Philosophie des Als Ob*.¹ In virtue of the continuity which through the fundamental form of synthesis makes itself felt in all life of the spirit, and especially within the intellectual life, analogies such as those mentioned become possible; naturally their logical value is highly various. It is, e.g., no fiction when a tangent is called a secant whose two points of intersection with a circle approach one another in the highest degree;

¹An interesting presentation and criticism of *Philosophie des Als Ob* has been given by JØRGEN FR. JØRGENSEN in the journal “Vor Tid”. 1914–15. p. 227–249.

for the synthesis can bring its elements together in a great many degrees. According to a modern tendency in geometry the tangent even always touches the circle in more than one point, so that the difference between secant and tangent falls away in the end.

7. Even between the two outermost limits of our cognition, the chaotic difference-series and the absolute identity-series, analogy is present. The only operation of thought they make possible is — apart from useless substitutions — counting. One can count how many absolute differences, or how many absolute identities, one knows. As we shall later see, arithmetic takes hold precisely where pure logic ends. Another analogy between them lies in this, that neither of the two series can be immediately given. Only through a strictly critical testing, which could not be shown to be exhaustive, could it be shown that there were no points of likeness at all, or no differences at all, within a series of subject-matters. The two series mark limits of our cognition which cannot be proved to be present in any single case whatever; only through arbitrary assertions could their validity be maintained. But these series serve indirectly to characterize human cognition, by exhibiting what would make it impossible from the first, or would be a conclusion admitting of no possible continuation.

One might say that epistemology has only the task of discussing the principles that make cognition possible, and that it must therefore first of all set up the principle which lies at the base of all formation of concepts and judgments and inferences whatever, namely the principle of identity. Cognition consists of judgments, one might say, and every judgment asserts a relation of identity (partial or absolute). The concept of identity would therefore seem to have to open the series of the categories. To this it must be answered that epistemology ought to investigate all possibilities. Every actual cognition is always only an approximation to identity, and all degrees of approximation must be investigated. Epistemology's proper subject-matter is the degrees of clash between identity and difference. Identity itself has only this significance, that it sets the advancing reduction of the differences as the great task. Identity is, after all, itself a relation. To strive after finding identity is therefore to seek to put one relation in the place of another. The concept of identity thus presupposes the concept of relation and the concept correlative with it, synthesis, and these become the first categories. Besides, identity can be defined only as a limiting case of likeness or of continuity, and these two concepts therefore go —

when we proceed analytically² — before the concept of identity.

When it is said that cognition consists of judgments, it must not be forgotten that the judgment's subject is always taken out of a whole connection, and that this taking-out becomes possible only through the difference there is between it and other subject-matters that are members of the same connection. Only by such a distinguishing does a subject-matter come to exist for us with such independence that closer determinations of it can be sought and found. As it has aptly been said, every concept has fringes or rents which show that it is properly a fragment. Only when this separating-out has been carried out can a judgment come into being. If every subject-matter drowned in, or were wholly fused with, the medium in which it exists from the first, no judgment would ever be passed. A relation of difference is both the logical and the psychological presupposition of a judgment's coming into being.

Naturally the principle of identity is presupposed in all logical thinking, also in the setting up of a doctrine of categories, and especially in the setting up of the categories synthesis and relation. Epistemology must of course build on the principles all science builds on; and since it is precisely these principles that epistemology wishes to bring forward and elucidate, it can to that extent be said to build on itself, or to move in a circle. Occasion has sometimes been taken from this to demonstrate the impossibility of an epistemology. So Schelling, Lotze and, in the most recent time, LEONARD NELSON (partly in his work *Über die sogenannte Erkenntnistheorie* 1908, whose first section is entitled *Die Unmöglichkeit der Erkenntnistheorie*, partly in a lecture at the congress of philosophers at Bologna in 1911). Nelson's train of thought is this: the criterion of cognition's truth must either be cognition or not be cognition. If it is cognition, then this cognition must itself require a criterion. And if it is not cognition, it must all the same be something that can be an object for cognition, and then we require a criterion for that cognition's being true. — This train of thought we know already from antiquity. It was brought forward on the sceptics' side against the Academics' attempt to set up a criterion of reality, when they asked by what criterion we can decide that the criterion of reality so set up is the right one!³ At the base of this scepticism, as of the modern objections to the possibility of epistemology, lies a confusion of the concepts of circularity and

²Cf. *Den menneskelige Tanke*, p. 177 f.

³See more on this in *Relation som Kategori*.

incompleteness. Epistemology's task must always be taken up anew. It must always be tested anew whether the principles which epistemology sets up as conditioning the validity of cognition coincide with the principles that lie at the base of the very attempt to set up the epistemology. If there is a difference here, the investigation must begin over again. A conclusion of these investigations is not to be thought. It may very well be that a new investigation of human thought-work can lead to the necessity of a new epistemology. Before Carneades too, and in modern times before Hume and Kant, men had an epistemology; but it showed itself that it had not been carried through, and that its fundamental significance had not been demonstrated.

Epistemology comes into being through the investigation of thought-work itself, that is to say, of the alteration which the application of certain principles brings about in respect of clarity and connection in our representations. If we find advancing clarity and connection, we say that epistemologically a step forward has been made, even if we perhaps come to strike out certain assumptions on which we formerly laid weight. The more numerous and the deeper the differences between the subject-matters that can be brought out, and the more we nevertheless succeed in finding grounded connection between them by the application of the principle of identity, the more we become convinced of that principle's validity. As we have seen, the principle of identity can be said to dawn already in the prescientific analogies, as well as in those series-formations which do not yet form a foundation for inferences. When, as in a regularly varying difference-series (e.g. the colour-series), each member lies fast, so that no interchange can take place, the principle of identity has come more into force than where the order is either quite arbitrary (as in the chaotic difference-series), or where the order is due to a wholly external point of view, e.g. a scheme for the memory (as in the indeterminately varying difference-series). And the history of the sciences shows us many stages in this respect before full clarity and certainty are attained.

8. In the analogies that play so great a role in prescientific thought-life there already stirs the energy which can later lead to scientific thinking. They ground the expectation with which men come to meet the subject-matters that present themselves to them, and which makes them not stand passive in the face of what observation offers. With a sanguine confidence in all emerging representations, wherever they may come from, the life of thought begins its course of development; indeed, even before free representations

appear, an analogy to this confidence declares itself in the involuntary propensity to immediate motor reaction, to which reflexes and instinctive actions testify. Life begins with a surplus of energy and has a tendency to spread and unfold itself. Before memory in the narrower sense of the word can form itself (with a clear consciousness that what is represented belongs to the past), expectation can be at work; and representations which are not bound up with any expectation, but whose content stands definitely as past, are held fast only when a higher development has been reached. Hume rightly laid weight on the need for expansion, for paying out the line, for continuing a train of thought one has once entered upon. But when he would derive from this tendency the belief in rational and causal principles, he overlooked that these principles come forward most clearly when expectations are disappointed or rejected, and when we are then compelled to seek an order and connection of another kind than the one our involuntary analogies and expectations had announced. Rationality and causality are norms which limit and dam in our involuntary course of thought, which in itself has a tendency to pursue certain ways of looking at things beyond all limits. The hypotheses which science on its way has shaped, but has then let fall again, testify that it is for a great part through disappointments that we work our way forward to truth. If disappointments had not come, we should at many points not have got beyond the prescientific stage of development; but the disappointments would not have come if the surplus energy had not fostered expectations which were involuntarily tested by our acting on them. Children's need to grasp and work upon all things leads them to make important experiences. They experiment involuntarily. The need for movement enters by degrees into the service of curiosity. In his autobiography the German surgeon Schleich relates that at five years old he had already been given a watch, but smashed it by knocking it with a stone; and when he was then scolded for it, he said: "Jungens müssen doch wissen, wa da'inn is!" This need to know what is "inside" stirs from the childish experimenting all the way up to the highest speculations, without there always stirring any intimation of how fine and how complete is that connection of things into which we seek to penetrate. That need is by degrees dammed in, as it is experienced that determinate laws rule for things, and that it is through the understanding of these that we may perhaps penetrate into what we call nature's interior.

In a spirited work (*De la contingence des lois de la nature*. 1875) ÉMILE BOUTROUX

developed the conception that the laws of nature signify essentially forms or channels for the unfolding of free forces — the river-bed for a stream that works its way forward. A step further went HENRI BERGSON, in demanding of philosophy that it should penetrate behind the mechanism of the laws of nature and make itself one with the force, the unfolding of life, which is constantly hampered and limited by the order of nature; and he found a foundation for this in an immediate intuition of our own inner unfolding of force (*élan vital*), in analogy with which we must then apprehend all other forms of existence than our own (Introduction à la Métaphysique. 1903). — Such attempts of thought are in reality due to a translation into metaphysical form of what psychology and epistemology teach. They are good examples of what we shall in a later connection go into somewhat more closely, that all metaphysics (cosmology) builds on analogy — here on analogy with the way in which we see, in the history of science, the human impulse to expansion and human impatience clash with the fixed order of things which science works to find. Of special interest here is a chapter in ERNST MACH's work *Erkenntnis und Irrtum* (1905). The chapter is headed "Sinn und Wert der Naturgesetze" and seeks to show that the laws of nature come into being for us as limitations which, under the guidance of experience, we set to our expectations, and that through such advancing limitation there takes place an ever closer adaptation of thought to facts. Even daily experience shows us a constant damming in of the involuntary life of thought.

9. Just as the cognition of laws is a limitation of the freedom to form connections of representations, so the formation of concepts is a limitation of the freedom to shape the single representations. We form concepts in order to be sure that a representation which emerges anew in our train of thought has the same content as when it appeared formerly; only thereby can it enter as a fixed member in a series of thoughts and stand in clear and determinate relations to the other members of the series. If it is merely a matter of holding fast what is peculiar to a subject-matter in a quite determinate situation, this is attained by what one might call a concrete individual concept. Thus "Napoleon at Waterloo" is a concrete individual concept, when we have noted Napoleon's whole conduct on that occasion. If the concept is to hold for all sides of a subject-matter's essence and for its appearance in all different situations — e.g. for Napoleon according to his whole character and course of life — we get a typical individual concept. Each single situation or each single action then serves only as an example of this concept.

And in a similar way it is with general concepts, which are to hold for a whole group of subject-matters. The “common” properties will here, just as little as with the typical individual concepts, be sufficient to be the content of a concept by itself. The “common” cannot be thought by itself alone. More or less distinctly, on the other hand, a single subject-matter stands as an example of the whole series; or one makes use of a scheme or a word which, if it be needed, can be replaced by an example.

Between the different examples themselves there will not be complete identity in all respects. It is some point of likeness (perhaps of identity) that leads one to set subject-matters together as examples of the same concept. An Indian who saw a book open for the first time called it an oyster. That was already an attempt at a formation of a concept. When men early spoke of feet both in animals and in men, and later extended the concept to tables, chairs, mountains and so on, that too was an attempt at the formation of a concept. But not even in scientific formation of concepts do the single examples completely cover one another. It is by the help of analogy that one goes from example to example within a concept’s extension. Under the concept of the conic section belong circle, ellipse, parabola, hyperbola, and the relation between these subject-matters is not identity but analogy. Only from a determinate point of view, which the definition of the conic section states in words, can they be regarded as identical, in so far as any one of them can in principle be used as an example just as well as the others. The very definition of the conic section (as a figure that arises by a plane section through a circular cone) requires a construction which leads to one or another of the examples adduced. It is the relation between properties or parts in a series of subject-matters that makes these subject-matters examples of the same concept. With the concept of colour there is provisionally only qualitative likeness between the single examples, and a concept proper is got only by keeping (as with the conic sections) to the way in which they arise. Each single colour-quality stands to a certain degree of refraction of light as the other colours stand to certain other degrees of refraction. For brevity’s sake (when one has not time to bring forward examples) words or signs (schemes) can express the concept. Thus letters in arithmetic are a scheme for the concept of magnitude, whose examples are had in the members of the number-series. “Mathematicians”, says PASCAL (*De l’esprit géométrique*), “and all who proceed methodically, give things names only in order to shorten the course of thought”. The word or the scheme is to give directions for the

construction of an example. Pascal requires that for safety's sake one should try to put what is defined in the place of the definition. But it must be remarked on this that, since "what is defined" comes forward in different shapes, this substitution can never become absolute. No example contains merely the features the definition states. Each single example requires other examples, and it is the analogy between them that is the concept's proper content. The concept's significance consists in its being able to set a series of thoughts going. GOBLOT therefore says that a concept is nothing but a possibility of a series of judgments (*une virtualité, une possibilité indéfinie de jugements*).⁴ He therefore begins his exposition of logic with the judgments and goes from there over to concepts and inferences. Earlier still, SIGWART had maintained the priority of judgments in relation to the concept. But the matter is surely more involved than the two excellent logicians present it. There is a constant reciprocal action between concept and judgment. The formation of concepts itself takes place through judgments, but the judgments are clear only when the concepts connected in them are each determinate. The history of thought shows us how the starting-point is taken now in the single concepts, now in the judgment. In epistemology this comes forward in the relation between fundamental concepts (categories) and fundamental propositions (principles).⁵

There is something in our thinking (and in our consciousness in general) which never comes forward for immediate observation, but which must be presupposed or, if one will, constructed, because what immediate observation exhibits is otherwise unintelligible. Thus synthesis, the fundamental form of conscious life and the first fundamental concept of all cognition, is not itself an observable element. Nor can our willing as such be immediately observed.⁶ And so it goes with all inner and outer activity: we observe its results and infer to the energy to which the work is due. The activity of thought itself is not observed. Every single thought is a member in a course of thought, and we infer to thought's energy from those alterations in the course of thought which are independent of momentary observations (except in so far as the alteration consists in such observations being brought into determinate connection with the previously given content of consciousness). Everywhere we observe only succession, not the activity which stretches through all the members of the succession and which we often have

⁴*Traité de Logique*. 1918. p. 87.

⁵Cf. *Den menneskelige Tanke*. § 51; 57; 106.

⁶*Psykologi*⁶ p. 177 f. (Forkortet Udgave p. 33 f.; 84 f.)

laboriously to trace out. Here Hume's criticism still holds good, in the domain of the human sciences as in that of natural science. His violent isolation of the single members in the course of thought led indirectly to the right apprehension of the concept of activity in all domains. But it is Leibniz who has directly indicated the right application of concepts such as force, energy and working, in maintaining that before there can be talk of force a lawlike connection between changing states must have been exhibited.⁷

Thinking proper is distinguished from the ordinary course of representations only by being directed towards the conditions for solving a task, towards means and ways of reaching a goal. It consists in an analysis of the connection within which the task or the goal is assumed to have its place, and then seeks to find a law among the elements to which the analysis leads. A simple example is presented by orientation when one has lost one's way. The different directions are tried and compared. If, on the other hand, one wanders without a determinate goal, one strikes into the first road that offers. Already HOBBS — who so often has to be the scapegoat in those criticisms of "association psychology" of which people seem unable to tire — saw this, when he declares that it is through constantly holding fast to a goal (*frequens ad finem respectio*. De Corpore XXV, 9) that we learn to find order and connection.

10. The concepts of analogy and of inference from analogy were first brought out with clarity by Aristotle. Whereas for Plato, even after he had had his last single combat with the Eleatics, it stood as the great goal to lead all cognition back to one Idea, Aristotle is convinced that we end in a plurality of concepts (categories), between which not identity but only analogy holds. Between the different domains in which thinking works, analogies too can perhaps be exhibited; thus Aristotle holds that the relation between matter (possibility) and form (actuality) recurs everywhere — but this relation itself takes on a different character in the different domains (that is to say, within the different sciences).

As regards the conducting of proof, Aristotle distinguishes between deduction and induction. By deductive inference he understands the syllogistic bringing of something under a genus-concept by the help of a species-concept as intermediate member. His doctrine of inference is thus built on a presupposed classification. By induction he understands a complete enumeration of all the subject-matters that come under a concept

⁷Den nyere Filosofis Historie.³ II p. 136 f.

(ἡ ἐπαγωγή διὰ πάντων. ed. Bekker p. 68 b). But there is also, he adds (p. 69 a), an inference from a single example, or rather from example to example. This inference thus goes neither, like deduction, from a whole to a part, nor, like induction, from all the parts to a whole, but it goes from one part to another part (ὡς μέρος πρὸς μέρος). It is distinguished from induction in that not all the parts, but only a single part, lies before us. Aristotle's designation for inference from analogy is Paradigm (παράδειγμα), which means example, but can also mean model. The single subject-matter can by inference from analogy be said to stand as a model, a type, in whose light a whole circle of subject-matters is seen, in that they present like relations between properties or parts among themselves. — The word ἀναλογία itself Aristotle uses mostly in the sense of mathematical proposition, identity of relation (ισότης λόγων), while for qualitative likeness of relation he frequently uses the indeterminate expression τὸ ἀνάλογον.

In more recent times it has been seen more and more clearly that there is no difference in principle between induction and analogy, since on the ground of experience one can never be sure of having all possible subject-matters of a certain kind given.⁸ Incomplete induction is then distinguished from inference from analogy only by having more examples to start out from. Naturally it is of importance to have as many examples as possible to start out from. But for modern methodology the main thing is whether, by an analysis of the subject-matters before us (one or several), a proposition can be found which can provisionally be set up so as then to be tested by seeing whether the inferences drawn from it find confirmation in new experiences. This method, which was first applied on a great scale by Galilei and Newton, is characteristic of modern natural science. Analogy and induction give, as KANT expresses it in his *Logic* (ed. 1801, p. 210), merely “logische Präsumtionen”. Strangely enough, he does not add that it may all the same often be of importance to test these surmises.

KANT himself saw clearly that there is a kind of inferring which is neither deduction, induction nor inference from analogy. He refers here to the principle by which mathematical thinking differs from formal logical thinking. In virtue of this principle an identically varying difference-series can be constructed, that is, a series within which the difference between the members following one another is always the same — where, accordingly,

⁸Except of course in such cases as those where it lies in the very definition of the concept that there must be a certain number of subject-matters. Cf. *Den menneskelige Tanke*. p. 172 f.

there is identity of difference. The simplest example of this is the number-series, which is built up by what Kant calls “Synthesis des Gleichartigen”. Besides in the number-series, the same principle appears — surely under the influence of the number-series as a model, and so in analogy with it — in the series of times, degrees and places, when (as with number) all special qualitative properties are purged away from these concepts. Judgments built on such series-formations (which may be called identical quality-series) Kant calls synthetic judgments *a priori*. In analogy with these series built on identity of difference, Kant has now constructed a concept of “possible experience” or “nature” as a scheme within which every subject-matter to which we are to be able to ascribe reality must find its place. We shall at a later point have occasion to discuss the significance of this construction.

While we are on the prehistory of the concept of analogy, it must be pointed out that in philosophical attempts to reach a world-view, a concluding cognition, the dispute in modern times stands between those who would reach a conclusion through inferences, that is, by the help of the principle of identity, and those who maintain that analogy is the last means of thought when it is a question of conclusion. It is the opposition between Plato and Aristotle repeating itself in new forms. On Plato’s side stand Spinoza and Hegel, on Aristotle’s side Leibniz, Schopenhauer, Lotze and, in the most recent time, Bergson. From SØREN KIERKEGAARD’S *Papirer*⁹ one sees that he was aware of the opposition between identity and analogy at the decisive points, in good agreement with the critical tendency of his epistemology. He had become clear that a systematic conclusion was impossible by the help of the principle of identity; and since induction and analogy permit no strict inferring, he came to the result that one always concludes (as one begins) with a leap — and it was surely a satisfaction to him to find an application here for one of his favourite concepts. But the problem is still more involved than he saw, and it is precisely the leap that poses it anew: the leap takes place in a curve, and so brings about a new continuity, though on another plane.

11. When a manifold of subject-matters lies before thinking, its first step will naturally be to order them in series whose members are as closely connected as possible.¹⁰ The most unfortunate relation would be that between all the subject-matters there subsist equally

⁹*Papirer*. Ed. P. A. Heiberg and V. Kuhr. V. p. 29.

¹⁰In what follows I make use of the attempt to set up and order the most important forms of series-formation which I have made in *Den menneskelige Tanke*. p. 162–173.

great differences, so that they can be put in any order whatever. As already shown above, it would be wrong to believe that such chaotic difference-series were peculiar to the primitive consciousness; this is characterized precisely by an involuntary interpretation, which lies at the base of the reflection it is able to institute. Only for a critical investigation, which is not content with any interpretation that offers, can it look as if the given were a chaos. As we have seen, the reality of a chaotic difference-series could not be proved. But the provisional assumption of a chaos can contain an invitation to closer investigation, and here analogy offers a starting-point. What would be common to the members of such a series would be precisely the circumstance that they are elements in a chaos; that holds for *A*, for *B*, for *C*, the whole series through. Herein lies the possibility of an inference in the Aristotelian second figure, one of the ways in which an analogy can be formulated. No positive inference can be derived from this figure, but there may be occasion to investigate whether after all there might not be other points of likeness between *A*, *B* and *C* than the one meagre point of belonging to a chaos. The second figure has in general the significance of setting a task. Already in this simple example, then, we see analogy's significance for thinking, as spurring, provoking and awakening. Analogy asks; but whether it answers is quite another question. And its very spurring effect presupposes that the participation has been dissolved, for by participation likeness at a single point is precisely made into likeness as regards the whole.

We do not get much further when the members lie fast and cannot without more ado change places, as when the order of succession is determined purely chronologically. It would have to be that the question were expressly raised why the subject-matters were experienced or remembered in just this determinate order. This too presupposes a wakeful reflection. A child who is taught the Mosaic story of creation might, e.g., ask why plants were created before animals, or how light could be created before sun and moon.

If we have a regularly varying difference-series, such as, e.g., the colour-series, then not only do the members lie fast, but a certain continuity also declares itself in the order of succession, in that the qualitative likeness to the neighbouring colour is always greater than that to a colour lying further off. And there can be analogous relations within the series, in that, e.g., violet may be said to stand to blue as orange to red. But above all the question can come up how this determinate continuous order is to be explained. Goethe

believed he could find an explanation by keeping to the qualitative analogies. But in spite of his protest, science has gone beyond this standpoint of quality by exhibiting an analogy between the degrees of refraction of light and the members of the quality-series. Instead of the analogy within the series, we get analogy between two series, of which the one is quantitative, the other qualitative. It will appear in what follows that an inference from analogy in this form is of decisive significance for the modern age's apprehension of existence.

If a series is symmetrical, that is, can be read in the reverse direction without the single members being interchanged or any member eliminated, there is a new possibility of analogy within the series. If (to use de Morgan's example) *A* is in conflict with *B*, and *B* with *C*, there is analogy between *A* and *C* with respect to their relation to *B*. The inference from analogy can be formulated in the second figure: *A* is an enemy of *B*, *C* is an enemy of *B*. The third figure can also be used: *B* is an enemy of *A* and is an enemy of *C*. But the analogy leads here to asking how the enmity between *A* and *C* more precisely stands. It may be that the common relation makes *A* and *C* friends. But it may also be that the common enmity is not enough to overcome their mutual indifference, or the enmity which for other reasons subsists between them.

With a progressive difference-series (that is, one in which each member stands to its predecessor as its successor stands to it) and with partial identity-series (that is, such as those in which each member is predicate to the preceding and subject to the following, or consequent of the preceding and ground of the following), not only are inferences from analogy like those already brought forward possible, but inferences can also be drawn by the simple elimination of intermediate members. A progressive difference-series is formed, e.g., by the numbers, the times, the degrees and the places, which we came to discuss above from another point of view. A partial identity-series is formed when predicates themselves become subjects for new predicates, and the consequents grounds for new consequents. The two kinds of series are characteristic of the formal sciences (logic and mathematics). But, as we have already seen, such series also acquire significance for purely qualitative difference-series. If it can be shown that a progressive difference-series or a partial identity-series within the circle of our subject-matters runs point for point parallel with a fixedly ordered quality-series, so that to a determinate member in the one series there answers a determinate member in the other, then from a

transition within the one series one can infer to a transition within the other. All exact empirical science rests on such a setting together of quantitative and qualitative series. Where such a setting together is possible, one can regard the quality-series as if they were purely logical or mathematical series. One can, e.g., regard the colour-series as if the difference between the colours consisted merely in the difference between the degrees of refraction of light; and one can regard unavoidable qualitative successions as if they were merely logical transitions from ground to consequent (in a partial identity-series), or mathematical transitions from one degree of refraction to another. But it is and remains analogies that are built on here, even though there has here been made the great advance, in relation to the inferences from analogy discussed above, that not merely is occasion given for questions, but determinate expectations and surmises are grounded — hypotheses, that is, are formed, which can be sought confirmed either by new members being looked for in the qualitative series, or by the quantitative series being shaped with greater exactness than before, perhaps even in such a way that a new quantitative series is constructed. Whereas certain irregularities in the orbit of Uranus could be explained from Newton's theory of gravitation, the irregularities in the orbit of Mercury cannot be explained from that theory, and EINSTEIN has then attempted an explanation from the electromagnetic theory, from which the theory of gravitation itself would then be derivable.¹¹

Whether the two series will continue to run parallel cannot be known with full certainty. However long a continuation of the quality-series may be, it does not vouch for it that new subject-matters may not arise which do not order themselves according to the rule stated in the quantitative series. Already LEIBNIZ saw that a complete verification is an impossibility. "A hypothesis' success", he says,¹² "does not prove its truth". To the complete confirmation of a hypothesis there would belong what Leibniz calls a complete reversal (*un parfait retour*) of the train of thought, that is, a demonstration that from the quality-series one came back with necessity to this determinate quantity-series and to no other. One would have to be able to reverse the inference and make its earlier premisses the conclusion. An objectively correct inference can, after all, be drawn from wrong premisses, and science's first (or last) presuppositions are therefore difficult to establish.

¹¹EINSTEIN: *Über die spezielle und die allgemeine Relativitätstheorie*⁶, p. 70. — EDDINGTON: *Space, Time and Gravitation*. 1920. p. 123–125.

¹²*Nouveaux Essais*. IV, 17, 5. (Op. phil. ed. Erdtmann. p. 397.)

A relation of identity is in any case not present here, though a dogmatic tendency leads again and again to assuming one, in order to be able to rest in a First (or a Last). Quality simply cannot be derived from quantity, even though the study of the quality-series can lead to the setting up of proportionally varying quantity-series. Quantity is itself a quality which (as with number, time, degree, place) lends itself to the setting up of progressive difference-series. Science aims everywhere, where it can be done, at forming such series, in order then by their help to find lawfulness within other series whose qualitative character cannot be reduced in that way. We shall later come to discuss some particular examples of the great analogy on which all modern exact empirical science rests, often without being conscious of it itself. We shall also come to consider more closely Kant's great merit of clarity on this point, as well as the misgiving that still remains in the face of his theory. Besides Kant, it is especially the famous physicist Clerk Maxwell (in philosophical respects a disciple of Kant at second hand) who has maintained the significance of the concept of analogy in this connection. To him Ernst Mach has on essential points attached himself. —

In conclusion it is to be remarked that even where absolute identity-series can be formed, we meet analogy as a presupposition. It was the Eleatic ideal to reduce everything to identity. If that could be reached, all existence's fundamental subject-matters could be presented in the series $A = B = C = D$ and so on. But (as I have shown in detail in *Den menneskelige Tanke* p. 177–201) every absolute identity between real subject-matters presupposes a determinate point of view in relation to which they can be regarded as identical, and so substituted for one another. From the point of view which the law for a group of subject-matters states, these subject-matters can be regarded as identical, though they may have many other properties than those the law concerns. It is the same relation as that between examples of the same concept.

Chapter III

Analogy between Functions of Cognition

Chapter IV

Analogy between Domains of Cognition

16. Logic and Arithmetic

17. Arithmetic and Geometry

18. Formal and Real Science

19. Natural Science and the Human Sciences

20. Ethics and Theoretical Science

21. World-view and Science

Chapter V

Poetic and Religious Symbolism